Paolo Nessim

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Education

- **Montgomery College** — Germantown, MD
  Associate of Arts in Computer Science; GPA: 3.58/4.0 — *Expected May 2027*
  - Dean's List; Phi Theta Kappa Honor Society; Southern Management Leadership Program (SMLP) Scholar
  - Planned transfer: Univ. of Maryland, B.S. Cyber-Physical Systems Engineering, Fall 2027 (expected May 2029)

Experience

- **University of Maryland, A. James Clark School of Engineering** — College Park, MD
  Research Intern, Cyber-Physical Systems Engineering (CPSE) — *Aug 2025 – Present*
  - Built a real-time computer-vision pipeline (Python, OpenCV) that detects and tracks physical chessboard state for the Magic Gambit v2 robotic chess system, driving robotic-arm motion planning from live piece movement.
  - Designed the system's custom ESP32 (WROOM-32E) controller PCB: schematic capture with zero ERC errors, USB-C programming with CH340C auto-reset, ESD protection, and a TPS61022 boost power chain, sourced against the LCSC catalog for JLCPCB fabrication.
  - Implemented an ESP-NOW wireless link to a RoArm-M2-S robotic arm and a real-time web interface for live board-state visualization and control.
- **University of Maryland, CPSE Summer Program** — College Park, MD
  Hardware/Software Engineer — *Jun 2025 – Aug 2025*
  - Developed "ICantSee," an assistive prototype combining live computer vision with haptic feedback so visually impaired users can sense obstacles in real time; selected from the cohort to continue as a research intern.
- **NASA L'SPACE Mission Concept Academy** — Remote
  Computer Engineering Participant — *Jan 2026 – May 2026*
  - Designed a space mission concept through Preliminary Design Review (PDR) on a multidisciplinary team, producing NASA-standard engineering documentation across systems, science, and operations sub-teams.
- **Nessimworks, Web Development & Software Services** — Germantown, MD
  Full-Stack Developer & UI Designer — *Jan 2023 – Present*
  - Design, ship, and maintain production websites and web apps for small-business and nonprofit clients (React, Next.js), owning requirements, delivery, and deployment end to end.

Projects

- **Snipprompt — Chrome Extension SaaS**: Prompt-snippet manager on the Chrome Web Store with paying subscribers ($9/mo Pro); Plasmo (MV3), Next.js, Better Auth, Stripe, Neon Postgres + Drizzle. Built and shipped solo.
- **Beacon — Self-Hosted Error Tracking Platform**: Lightweight Sentry alternative: Python ingestion core and REST API (49 tests), React dashboard, Python and Node.js client SDKs, Docker builds, alerting.
- **OpenChess — Smart Chessboard Hardware**: Custom ESP32 PCB (KiCad) with 64 hall-effect sensors for piece detection, WS2812 LED matrix, OLED, and BLE app; C++ firmware (PlatformIO); plays online via Lichess.
- **Satgaze — Real-Time 3D Satellite Tracker**: Live positions of 24,000+ satellites via SGP4/TLE orbital propagation in the browser (Next.js, TypeScript, Three.js/WebGL).
- **YouTube AI Extension — Open Source**: Chat-with-any-video Chrome extension (React, Plasmo, OpenAI API); 665 GitHub stars.

Leadership & Activities

- Co-Lead, AI Project Team, Raptors Who Code (Montgomery College), Feb 2024 – Present; MC AI Club member.

Skills

- **Languages**: Python, C/C++ (embedded), TypeScript/JavaScript, Java, SQL
  Hardware & Embedded: KiCad schematic & PCB design, ESP32 (ESP-IDF/PlatformIO), ESP-NOW, BLE, Arduino, sensor integration (hall-effect, haptic), oscilloscopes, circuit prototyping, Fusion 360
  Software: OpenCV, ROS, React, Next.js, Node.js, Docker, PostgreSQL, Git, Linux